

Test 33: L-scaling χ^2 (10 $\times$ 20 $\times$ 30 $\times$ 50 $\times$ 80 $\times$ 112)

Verdict: PASS | Generated: 2026-09-02 21:32:46 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors L $\geq$ 50, extended points L=80/112 (user spec 2026-08-29) check the GUE plateau; W_{eff} capped

METHOD: Statistical finite-size trend: monotone approach to 0.594-0.599; monograph anchors L $\geq$ 50, extended points L=80/112 (user spec 2026-08-29) check the GUE plateau; W_{eff} capped

Test 33 HARDCORE pass 2: $n \geq 5$ per L, weighted GUE-plateau χ^2 over L $\geq$ 50, L capped at 112 (RAM honesty).

WHY W (DISORDER) IS CAPPED AT $W_{\text{max}} \geq 1.0 \geq$ disorder χ^2 vortex motion χ^2 statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k W / (r_{ik}^2 + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy $q \bullet W$) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

$\chi^2 q \bullet W \geq t$ (e.g. $W=4$, $q=0.3 \Rightarrow V \approx 1.2$ next to a core): the disorder landscape dominates the vortex motion χ^2 vortices pin/steer the eigenstates, states localize on the potential relief, and χ^2 drifts from GUE (0.5992) toward Poisson (0.3863) by an essentially random amount. A strongly disordered, spinning vortex system can therefore emit ALMOST ANY statistics χ^2 the numbers depend on the effective rotation speed (W/t) and on the twisting/vortex arrangement (N_v , layout, charges) of that realization, not on the AB physics.

$\chi^2 q \bullet W \geq t$ ($W \geq 1$): the Aharonov-Bohm phases dominate the hopping, states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at $W_{\text{eff}} = \min(\text{cfg.ab_}W, W_{\text{max}} = 1.0)$: they certify the PHASE physics, not the disorder physics.

Test 33 RAM gate: L=112 skipped (needs $\geq$ 10.6 GB peak, free $\geq$ 4.5 GB).

Extended L-scan to L=80 (user spec 2026-08-29); $W_{\text{eff}}=1.00$; verdict $q=0.30$; trend 0.434 $\geq$ 0.556 $\geq$ 0.593 $\geq$ 0.6 $\geq$ 0.6

Test 33 HARDCORE plateau ($q=0.30$): weighted $\chi^2=0.6004$ vs GUE ($|\chi^2|=0.0012$, in band), $\chi^2/\text{dof}=0.01$.

Test 33b (hardcore): plateau weighted $\chi^2=0.6004$, $\chi^2/\text{dof}=0.01 \geq$ PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' χ^2 animated reveal of every plot of this test

Result

Plots

Independent verification guide

Reproduce with: `julia ab_cloud_v19.jl --test 33 --no-two-pass`

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

FULL COMPUTATION LOG

[21:18:17.797] [+27013.080s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[21:18:18.357] [+27013.639s] calc: ?r? = 0.629404 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:19.481] [+27014.763s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[21:18:20.034] [+27015.316s] calc: ?r? = 0.509721 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:21.171] [+27016.453s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[21:18:21.721] [+27017.004s] calc: ?r? = 0.574634 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:22.850] [+27018.132s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[21:18:23.385] [+27018.667s] calc: ?r? = 0.557955 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:24.495] [+27019.777s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[21:18:25.049] [+27020.331s] calc: ?r? = 0.486070 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:26.189] [+27021.472s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[21:18:26.787] [+27022.069s] calc: ?r? = 0.593495 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:27.998] [+27023.280s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[21:18:28.604] [+27023.886s] calc: ?r? = 0.596397 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:29.710] [+27024.992s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[21:18:30.274] [+27025.556s] calc: ?r? = 0.582880 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:31.321] [+27026.603s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[21:18:31.864] [+27027.146s] calc: ?r? = 0.588530 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:32.818] [+27028.100s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, torus, 12 vortices, model=:monumental

[21:18:33.332] [+27028.614s] calc: ?r? = 0.615185 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:34.335] [+27029.618s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[21:18:35.120] [+27030.402s] calc: ?r? = 0.625316 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:36.129] [+27031.411s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[21:18:36.862] [+27032.144s] calc: ?r? = 0.610951 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:37.869] [+27033.151s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[21:18:38.593] [+27033.875s] calc: ?r? = 0.600825 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:39.581] [+27034.863s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[21:18:40.302] [+27035.584s] calc: ?r? = 0.592813 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:41.270] [+27036.552s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, torus, 26 vortices, model=:monumental

[21:18:42.004] [+27037.286s] calc: ?r? = 0.605293 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:43.019] [+27038.302s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, torus, 70 vortices, model=:monumental

[21:18:48.015] [+27043.297s] calc: ?r? = 0.597483 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:49.075] [+27044.357s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, torus, 70 vortices, model=:monumental

[21:18:54.049] [+27049.331s] calc: ?r? = 0.594415 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:18:55.096] [+27050.378s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, torus, 70 vortices, model=:monumental

[21:19:00.056] [+27055.338s] calc: ?r? = 0.606851 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:19:01.083] [+27056.365s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, torus, 70 vortices, model=:monumental

[21:19:06.087] [+27061.370s] calc: ?r? = 0.574458 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:19:07.153] [+27062.436s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, torus, 70 vortices, model=:monumental

[21:19:12.078] [+27067.360s] calc: ?r? = 0.594551 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:19:13.463] [+27068.745s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, torus, 178 vortices, model=:monumental

[21:20:25.768] [+27141.050s] calc: ?r? = 0.598524 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:20:27.289] [+27142.571s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, torus, 178 vortices, model=:monumental

[21:21:39.539] [+27214.821s] calc: ?r? = 0.593096 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:21:41.084] [+27216.366s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, torus, 178 vortices, model=:monumental

[21:22:54.656] [+27289.938s] calc: ?r? = 0.598961 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:22:56.054] [+27291.337s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, torus, 178 vortices, model=:monumental

[21:24:11.640] [+27366.922s] calc: ?r? = 0.602712 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:24:13.429] [+27368.711s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, torus, 178 vortices, model=:monumental

[21:25:27.096] [+27442.378s] calc: ?r? = 0.600199 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:28.354] [+27443.636s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental

[21:25:28.912] [+27444.194s] calc: ?r? = 0.512863 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:30.090] [+27445.372s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental

[21:25:30.645] [+27445.928s] calc: ?r? = 0.408227 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:31.782] [+27447.064s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental

[21:25:32.334] [+27447.616s] calc: ?r? = 0.434605 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:33.453] [+27448.735s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental

[21:25:34.001] [+27449.283s] calc: ?r? = 0.444497 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:35.115] [+27450.397s] build_ab_cloud_hamiltonian: 10x10 (N=100), ?=0.5000, t=1.00, W=1.00, open, 4 vortices, model=:monumental

[21:25:35.649] [+27450.931s] calc: ?r? = 0.371493 (n_r=59, n_eigs=61) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:36.772] [+27452.054s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[21:25:37.377] [+27452.659s] calc: ?r? = 0.552303 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:38.506] [+27453.789s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[21:25:39.119] [+27454.401s] calc: ?r? = 0.568758 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:40.260] [+27455.542s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[21:25:40.847] [+27456.129s] calc: ?r? = 0.525701 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:41.974] [+27457.256s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[21:25:42.526] [+27457.808s] calc: ?r? = 0.549972 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:43.537] [+27458.819s] build_ab_cloud_hamiltonian: 20x20 (N=400), ?=0.5000, t=1.00, W=1.00, open, 12 vortices, model=:monumental

[21:25:44.100] [+27459.382s] calc: ?r? = 0.585633 (n_r=239, n_eigs=241) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:45.054] [+27460.336s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[21:25:45.772] [+27461.054s] calc: ?r? = 0.578164 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:46.753] [+27462.035s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[21:25:47.487] [+27462.769s] calc: ?r? = 0.588849 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:48.476] [+27463.758s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[21:25:49.202] [+27464.484s] calc: ?r? = 0.583585 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:50.205] [+27465.487s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[21:25:50.928] [+27466.210s] calc: ?r? = 0.601887 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:51.924] [+27467.205s] build_ab_cloud_hamiltonian: 30x30 (N=900), ?=0.5000, t=1.00, W=1.00, open, 26 vortices, model=:monumental

[21:25:52.656] [+27467.938s] calc: ?r? = 0.614023 (n_r=539, n_eigs=541) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:53.720] [+27469.003s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[21:25:58.716] [+27473.998s] calc: ?r? = 0.606783 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:25:59.762] [+27475.044s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[21:26:04.716] [+27479.999s] calc: ?r? = 0.602431 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:26:05.772] [+27481.054s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[21:26:10.722] [+27486.005s] calc: ?r? = 0.597144 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:26:11.792] [+27487.074s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[21:26:16.739] [+27492.021s] calc: ?r? = 0.608952 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:26:17.808] [+27493.090s] build_ab_cloud_hamiltonian: 50x50 (N=2500), ?=0.5000, t=1.00, W=1.00, open, 70 vortices, model=:monumental

[21:26:22.798] [+27498.080s] calc: ?r? = 0.586730 (n_r=1499, n_eigs=1501) [refs: GUE=0.5996 Poisson=0.3863]

[21:26:24.162] [+27499.444s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[21:27:34.829] [+27570.111s] calc: ?r? = 0.597930 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:27:36.326] [+27571.608s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[21:28:48.328] [+27643.610s] calc: ?r? = 0.610283 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:28:49.848] [+27645.130s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[21:30:02.856] [+27718.138s] calc: ?r? = 0.595793 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:30:04.465] [+27719.747s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[21:31:23.977] [+27799.259s] calc: ?r? = 0.599605 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

[21:31:26.491] [+27801.773s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ?=0.5000, t=1.00, W=1.00, open, 178 vortices, model=:monumental

[21:32:42.881] [+27878.163s] calc: ?r? = 0.598340 (n_r=3839, n_eigs=3841) [refs: GUE=0.5996 Poisson=0.3863]

CONSOLE CAPTURE

??

TEST 33 (pass 2, HARDCORE): L-scaling ? deep validation

Harder machinery: up to 5 realizations per L, weighted

GUE-plateau ?? over L ? 50, no L beyond 112 (RAM honesty: L=128 ? 4.3 GB).

??

? RUNTIME: 2 q-rows ×up to n=5 realizations ×L ? {10,20,30,50,80,112} ? expect ? 27?81 min (−3×machine factor; RAM-gated rungs are skipped inside with a printed reason)

??

TEST 33: L-Scaling for AB-Cloud Lattice

L=10?0.487 (Poisson), L=20?0.562, L=30?0.595, L=50?0.594 (monograph),

L=80, L=112 ? GUE plateau check (extended range, user spec 2026-08-29)

??

WHY W (DISORDER) IS CAPPED AT W_max ? 1.0 ? disorder ? vortex motion ? statistics:

W is the diagonal on-site potential the vortices paint onto the lattice:

$$V_i = \sum_k q_k \cdot W / (r_{ik} \cdot N_{\cdot} + 1), \quad \text{hopping } t = 1.$$

The ratio (disorder energy q•W) / (kinetic energy t) decides what the

vortex cloud does and which spectral statistics come out:

? q•W ? t (e.g. W=4, q=0.3 ? V?1.2 next to a core): the disorder

landscape dominates the vortex motion ? vortices pin/steer the

eigenstates, states localize on the potential relief, and ?r? drifts

from GUE (0.5992) toward Poisson (0.3863) by an essentially random

amount. A strongly disordered, spinning vortex system can therefore

emit ALMOST ANY statistics ? the numbers depend on the effective

rotation speed (W/t) and on the twisting/vortex arrangement (Nv,

layout, charges) of that realization, not on the AB physics.

? q•W ? t (W ? 1): the Aharonov-Bohm phases dominate the hopping,

states stay delocalized and GUE (Wigner-Dyson) is reachable.

That is why these tests measure at W_eff = min(cfg.ab_W, W_max = 1.0):

they certify the PHASE physics, not the disorder physics.

? RAM GATE: L=112 needs ? 10.6 GB peak (matrix + LAPACK workspace), free ? 4.5 GB ? SKIPPED (a swap-thrashing eig is indistinguishable from a hang)

q=1.00 BC=:torus L=10 Nv=4 ?=0.50 W=1.00 ?r? = 0.5516 – 0.0252 (5 real, monograph 0.487)

q=1.00 BC=:torus L=20 Nv=12 ?=0.50 W=1.00 ?r? = 0.5953 – 0.0055 (5 real, monograph 0.562)

q=1.00 BC=:torus L=30 Nv=26 ?=0.50 W=1.00 ?r? = 0.6070 – 0.0054 (5 real, monograph 0.595)

q=1.00 BC=:torus L=50 Nv=70 ?=0.50 W=1.00 ?r? = 0.5936 – 0.0053 (5 real, monograph 0.594)

q=1.00 BC=:torus L=80 Nv=178 ?=0.50 W=1.00 ?r? = 0.5987 – 0.0016 (5 real, monograph 0.595*)

q=0.30 BC=:open L=10 Nv=4 ?=0.50 W=1.00 ?r? = 0.4343 – 0.0233 (5 real, monograph 0.487)

q=0.30 BC=:open L=20 Nv=12 ?=0.50 W=1.00 ?r? = 0.5565 – 0.0100 (5 real, monograph 0.562)

q=0.30 BC=:open L=30 Nv=26 ?=0.50 W=1.00 ?r? = 0.5933 – 0.0065 (5 real, monograph 0.595)

q=0.30 BC=:open L=50 Nv=70 ?=0.50 W=1.00 ?r? = 0.6004 – 0.0040 (5 real, monograph 0.594)

q=0.30 BC=:open L=80 Nv=178 ?=0.50 W=1.00 ?r? = 0.6004 – 0.0025 (5 real, monograph 0.595*)

Verdict run: q=0.30 (fractional AB phase ? reference standard, stable)

Trend (L=10?20?30?50?80): 0.434 ? 0.556 ? 0.593 ? 0.6 ? 0.6

Monotone increase: YES

Final L=80 close to GUE 0.595 (|?|<0.05): YES

(comparison) q=1.0 user setting: 0.552 ? 0.595 ? 0.607 ? 0.594 ? 0.599 (final 0.5987)

Test 33: L-scaling to L=80 0.434,0.556,0.593,0.6,0.6 ? PASS

??? HARDCORE GUE-plateau check (verdict q=0.3, L ? 50) ???

plateau points: L=50: 0.6004–0.0040, L=80: 0.6004–0.0025

weighted ?r? = 0.6004 vs GUE 0.5992 (|?|=0.0012) ? IN ? band –0.022

?? = 0.01, dof = 2, ??/dof = 0.01 ? consistent ? (? 2 consistent; diagnostic)

Test 33b (hardcore): plateau weighted ?r?=0.6004, ??/dof=0.01 ? PASS